

E-Commerce Health Check

Mini-Capstone Project — Data Science

Dataset: Olist Brazilian E-Commerce Public Dataset **Tools:** Python, Pandas, NumPy, SQL, Matplotlib

1. Project Overview

This project performs an E-Commerce Health Check using the Olist Brazilian E-Commerce Public Dataset. The analysis focuses on customer behavior, product revenue, order-value patterns, delivery performance, customer satisfaction, and order trends. Python, Pandas, NumPy, SQL, and Matplotlib were used to clean, analyze, and visualize the data.

2. Methodology

The dataset was loaded into Python using Pandas. The data was inspected for missing values and duplicates, and date columns were converted into datetime format. Multiple datasets were merged using common identifiers such as order_id, customer_id, and product_id. Order values were calculated using product price and freight value.

SQL was used to analyze top products by revenue, repeat customer rate, and monthly order trends. CTEs, JOINS, GROUP BY, and a window function were used in the SQL analysis. Exploratory Data Analysis was performed using descriptive statistics and visualizations. The relationship between delivery time and review score was analyzed using correlation. IQR was used for outlier detection, and a 95% confidence interval was calculated for the average order value.

3. Key Results

Metric	Result
Average Order Value	■160.58
Median Order Value	■105.29
95% Confidence Interval	■159.20 – ■161.95
Repeat Customer Rate	3.12%
Delivery–Review Correlation	-0.33
Weekday Orders	76,594
Weekend Orders	22,847
Order-Value Outliers	7,775
Highest Product Revenue	■63,885

4. Key Findings

- Low Repeat Customer Rate:** The repeat customer rate is only 3.12%, indicating that most customers placed only one order.
- Delivery Time and Customer Satisfaction:** The correlation between delivery time and review score is -0.33, indicating that longer delivery times are generally associated with lower review scores.
- Right-Skewed Order Values:** The average order value is ■160.58, while the median is ■105.29. This indicates that some high-value orders increase the average.
- Weekday Orders Dominate:** There were 76,594 weekday orders compared with 22,847 weekend orders, meaning approximately 77% of orders occurred on weekdays.
- High-Revenue Products:** The top product generated ■63,885 in product revenue, showing that some products contribute significantly to sales revenue.

5. Business Recommendation

The company should focus on improving customer retention and delivery performance. Since the repeat customer rate is only 3.12% and longer delivery times are associated with lower review scores, the business can introduce targeted offers for existing customers and improve delivery speed. The company should also monitor high-value products and

ensure their availability because a small number of products generate significant revenue.

6. Conclusion

The E-Commerce Health Check shows that the dataset contains a large number of weekday orders, a low repeat customer rate, and a right-skewed order-value distribution. The analysis also found a negative relationship between delivery time and review score. These findings can help the business focus on customer retention, delivery performance, and monitoring high-revenue products.